



ENGF0034: Design and Professional Skills

Introductory lecture

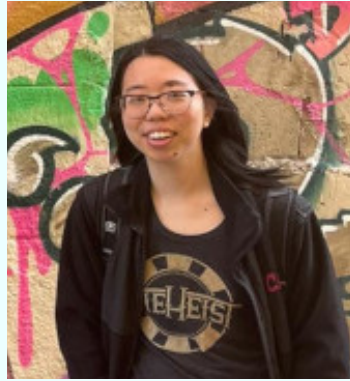
Prof. Martin Benning
30 September 2025



ENGF0034 2025/26 is a team effort



Chika Nweke



Yuzuko Nakamura



Vivek Ramachandran



Fiona Truscott



Sunny Bains

Teaching Assistants



Yiliu (Aiden) Li



Yiqian (Ken) Liu

And many more

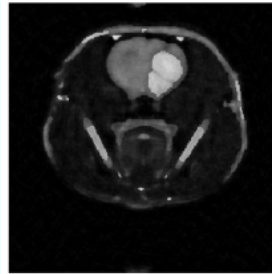
Who am I? Professor of Inverse Problems



Martin Benning

Areas of expertise:

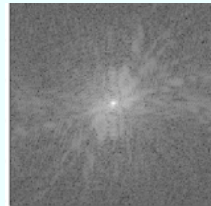
- Inverse & Ill-posed problems
- Regularisation theory
- Machine learning
- Optimisation
- Industrial & medical Imaging
- Image processing



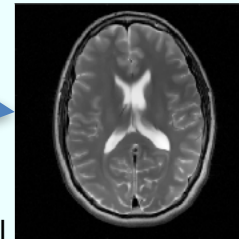
Joint reconstruction
& segmentation



Nonlinear spectral decomposition



Medical MRI



Transmission Electron Microscopy / Tomography

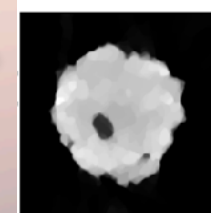
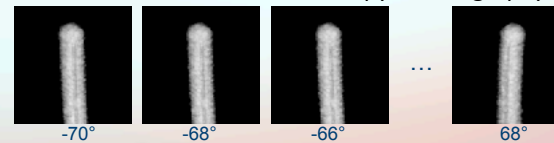
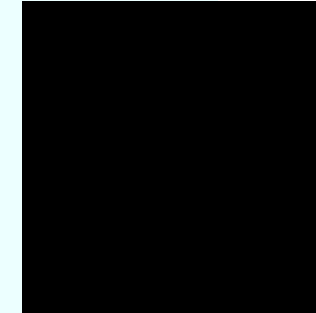
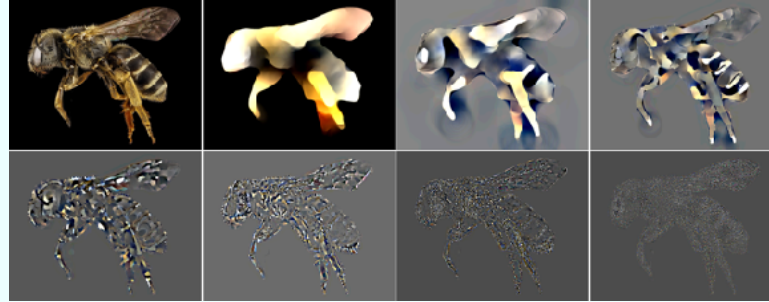
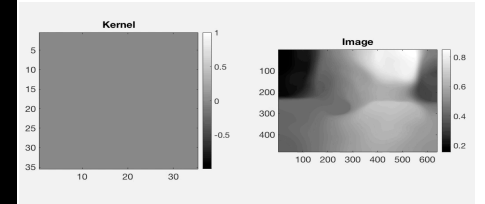


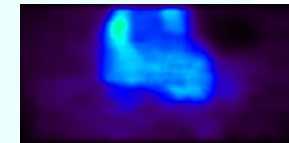
Image reconstruction



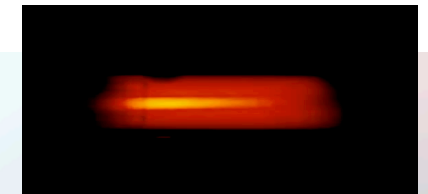
Blind image deconvolution



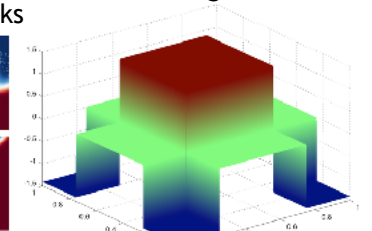
Medical PET



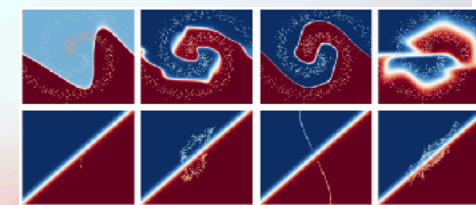
Industrial MRI



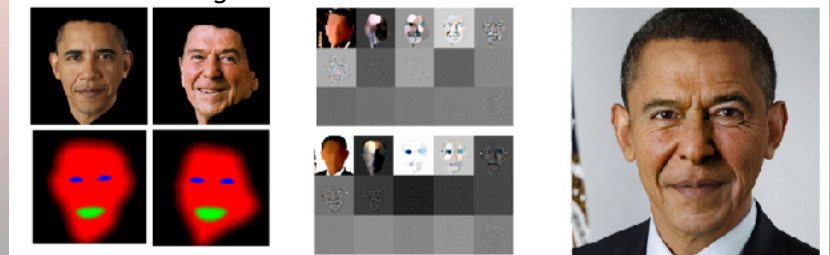
Nonlinear eigenfunctions



Classification with residual neural networks



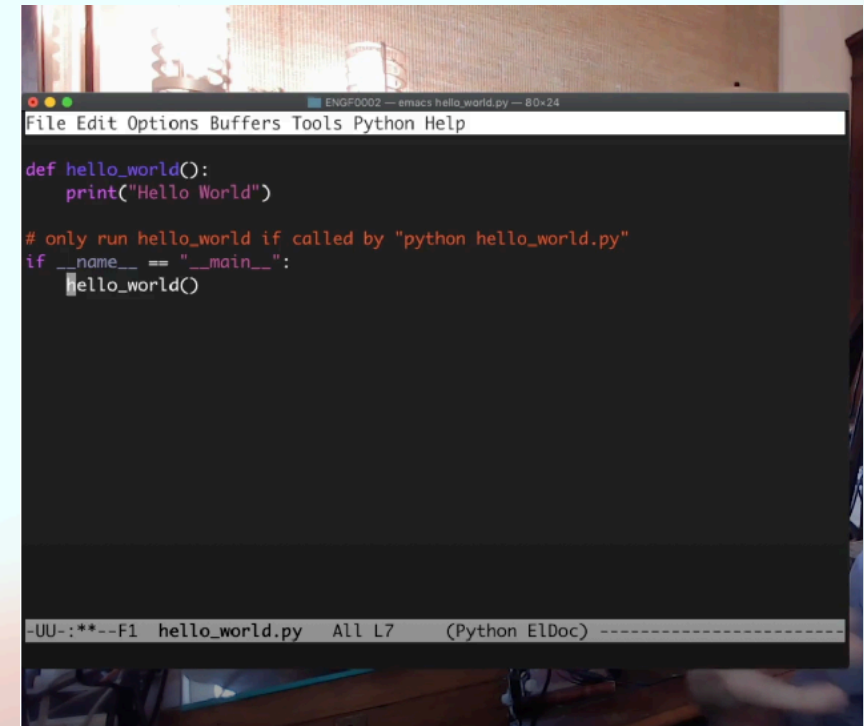
Nonlinear image fusion



Module approach

- Lectures (almost) every **Tuesday**
- Tutorials (almost) every **Friday**
- **Ask and answer questions** via the Moodle forum
- **Course materials** will be released via [git](#)
- **Assessments** will be released via [Github Classroom](#)
(or Moodle)
- **Drop-in clinics** will take place as extra support for those new to programming

Let us checkout the module [Moodle](#) & [git](#) pages



Drop-in clinics

Teaching Assistants



Yiliu (Aiden) Li



Yiqian (Ken) Liu

- Drop-in clinic Aiden & Ken (in-person): Thursday, 4-5 pm
G03, Henry Morley
- Drop-in clinic Aiden (in-person): Friday, 1-2 pm
G03, Henry Morley
- Drop-in clinic Ken (in-person): Wednesday 3-4 pm
G03, Henry Morley

(Starts next week!)

Assignments

- Week 1 - 2: Introductory programming assignment
- Week 2 - 3: Debug the bomber game
- Week 3 - 5: Debug the frogger game
- Week 5 - 7: Write a tetris autoplayer (20%)
- Week 7 - 9: Write pacman protocol specification (15%)
- Week 7 - March 22 in Term 2: Ethics report (15%)
- Term 2 Reading Week: Presentations (10%)
- + Scenario assessments in Term 2 (remaining 40%)

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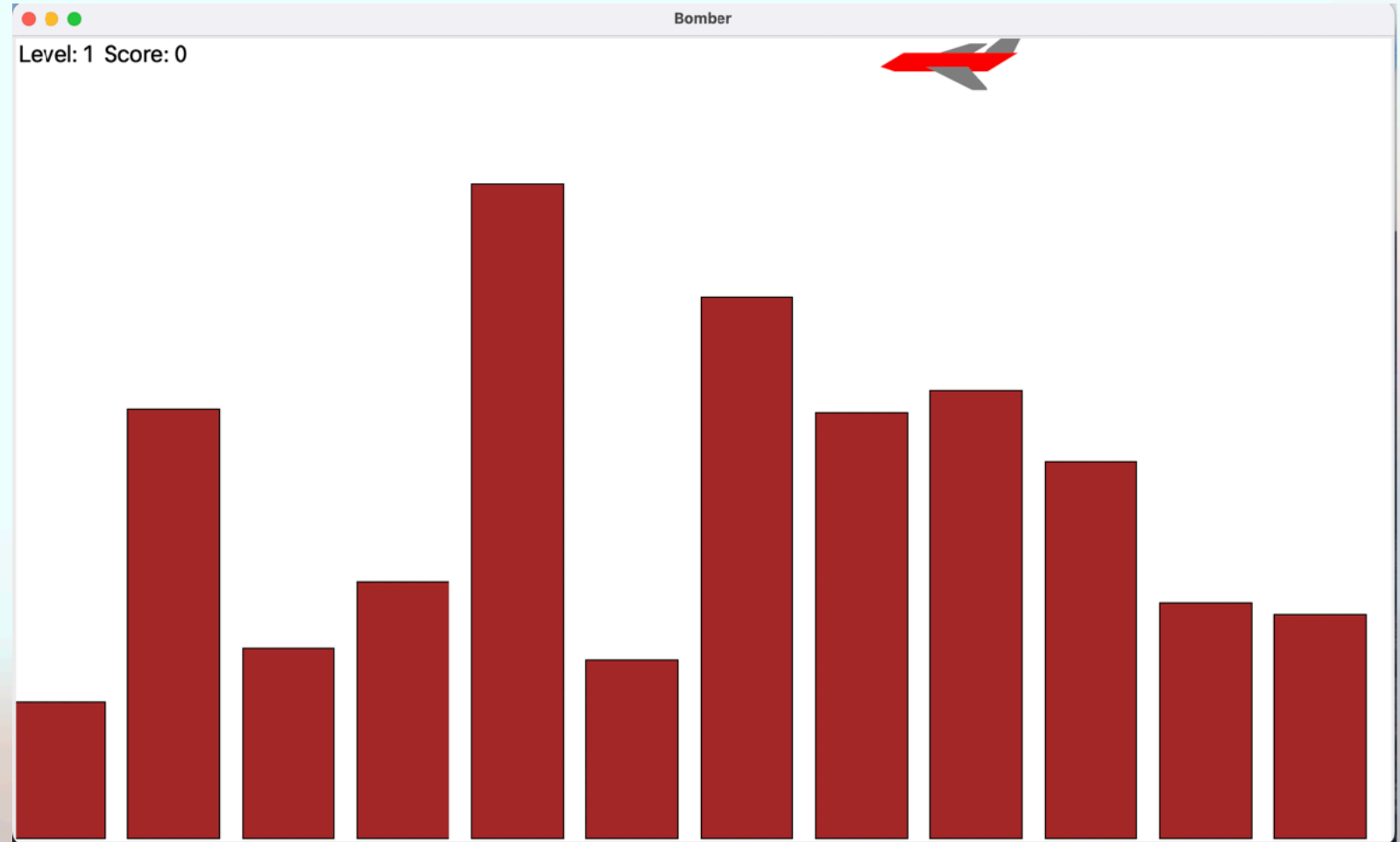
Do not plan time away
during reading week!

Assignments

Assignment 2: (Canyon) Bomber

Classic Atari Arcade Game from 1977

Task: Find five bugs that make the game unplayable, and fix them!

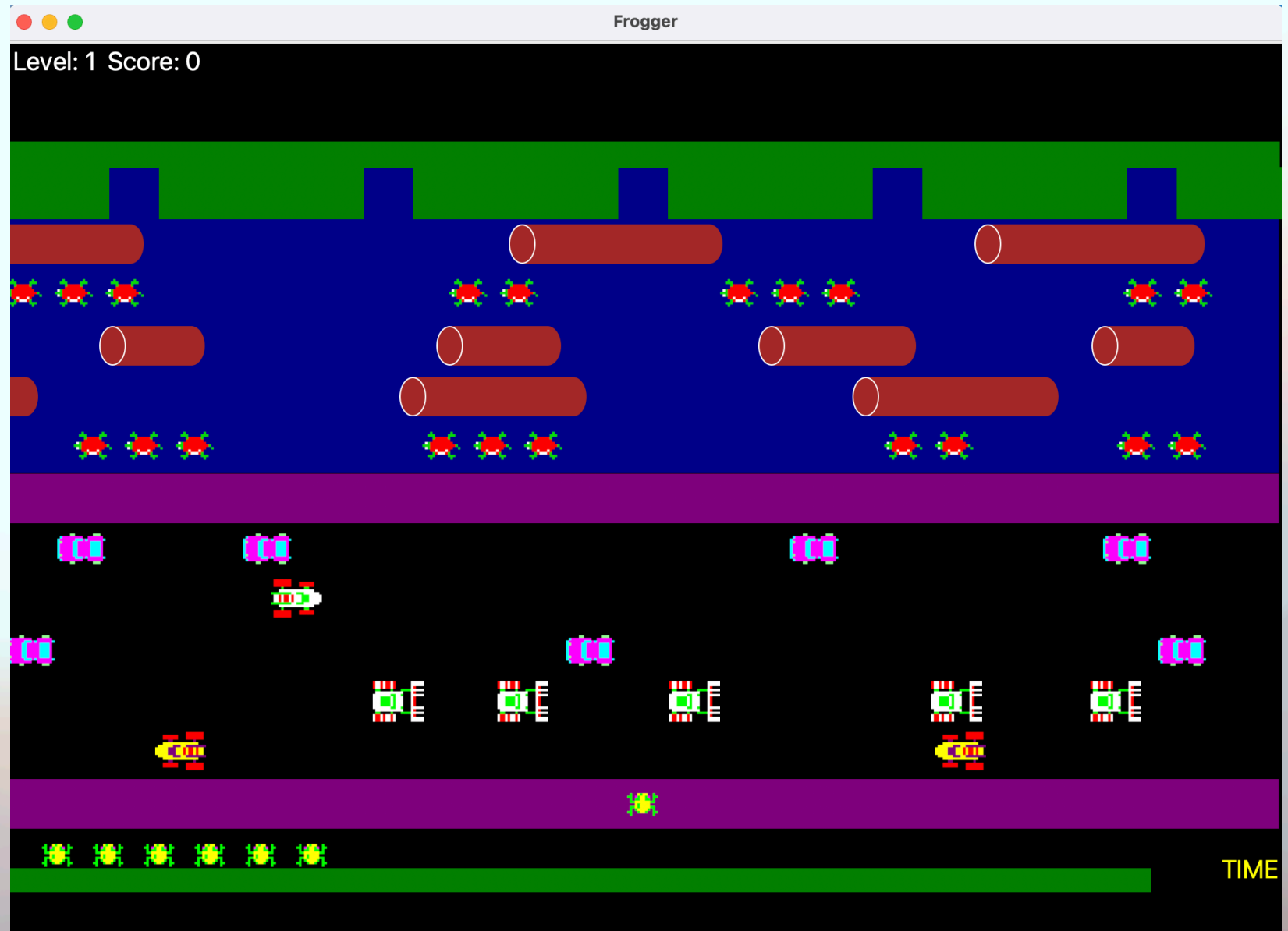


Assignments

Assignment 3: Frogger

Classic Konami/Sega Arcade
Game from 1981

Task: Find five bugs that make
the game unplayable, and fix
them!



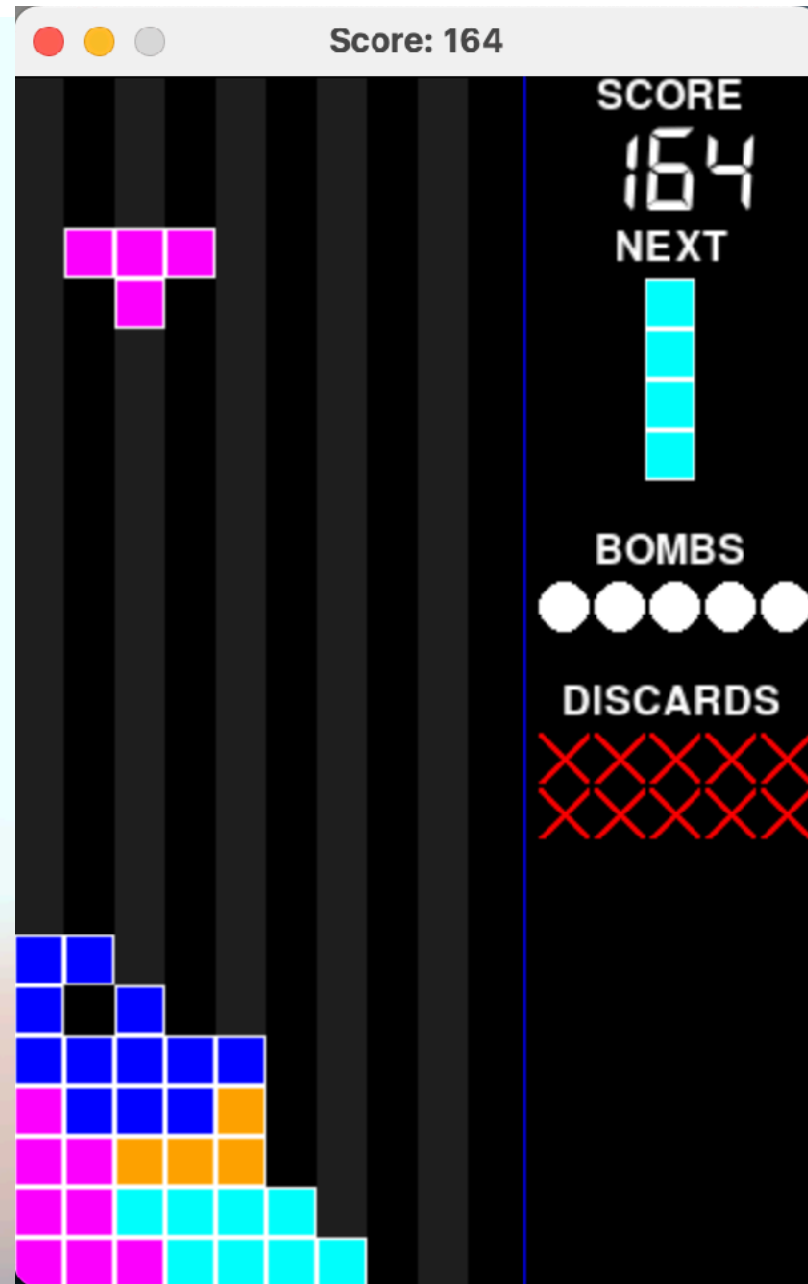
Assignments

Assignment 4: Tetris autoplayer

Puzzle video game created by
Alexey Pajitnov in 1985

Game Boy version by Nintendo
popularised Tetris in 1989

Task: write a program that
plays Tetris and compete with
your peers



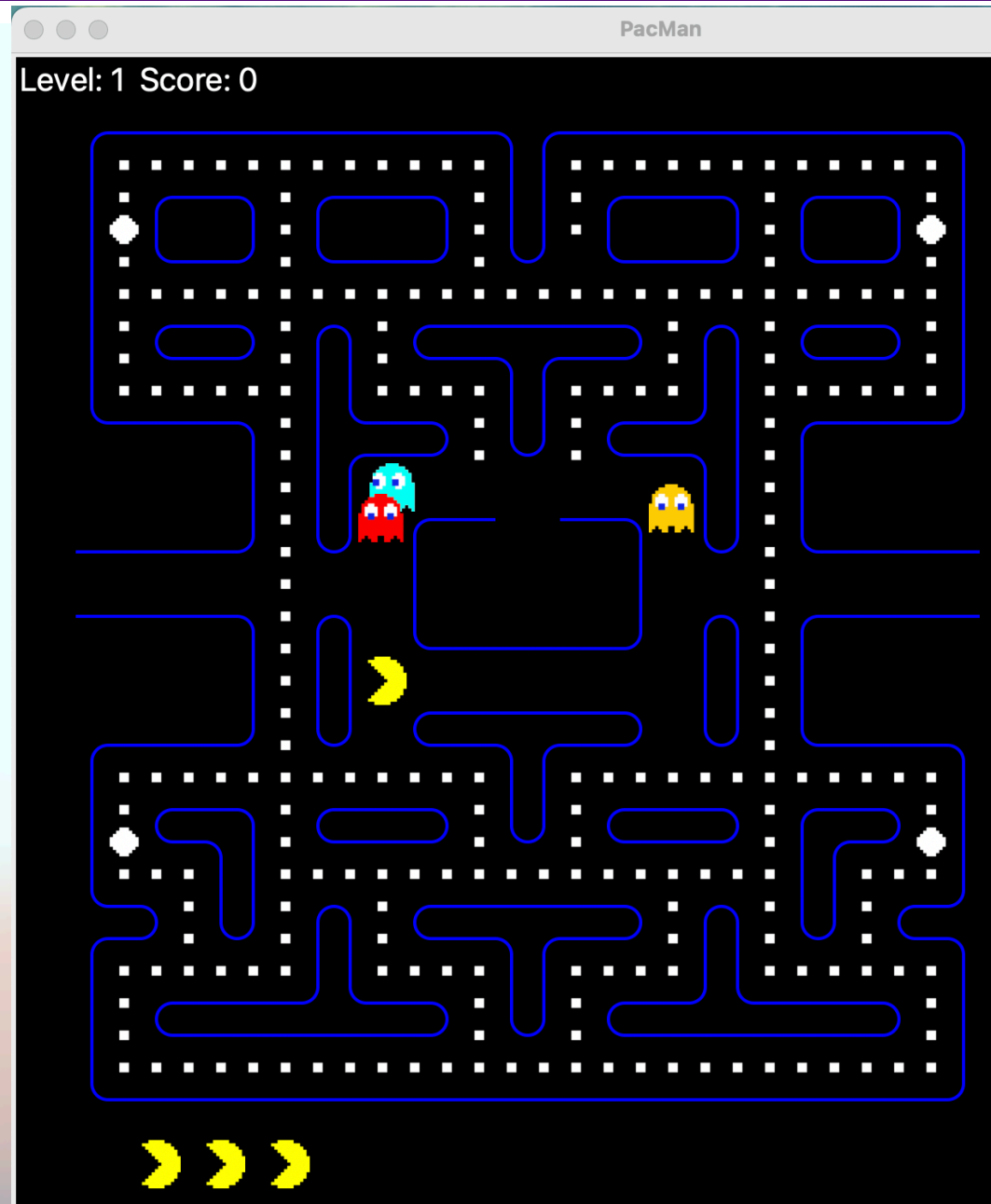
Assignments

Assignment 5: Write specification for Pacman game

1980 Maze video game

Special network version of the game

Task: write a better
network protocol
specification for this game

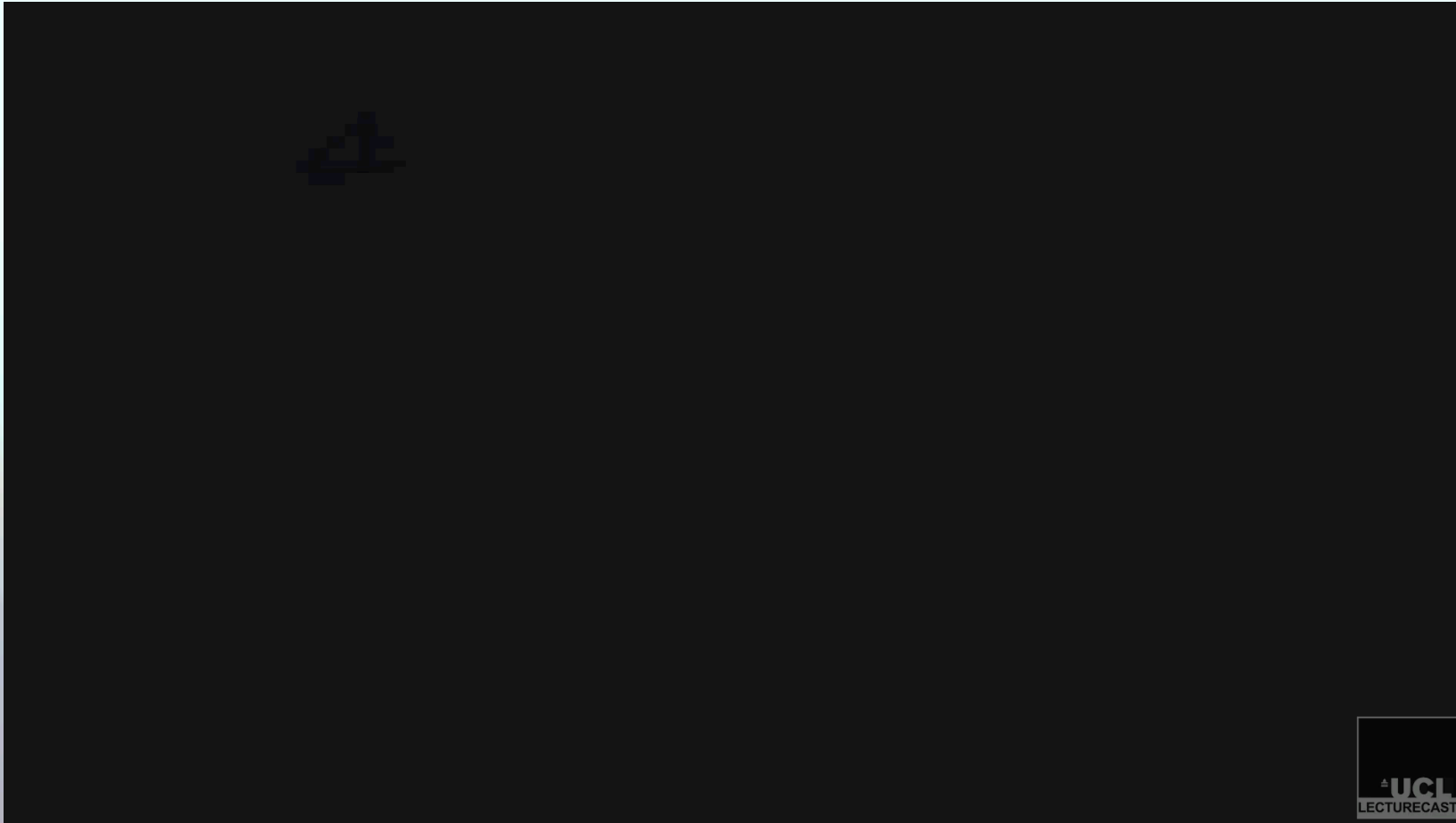


Other remarks

- For assessments 1 - 3 communicate and work with your peers
- If you have questions, it is likely your peers will have similar questions ➡ ask via Moodle forum, **not** via e-mail
- Do not only use forum when you have questions; help your peers and answer their questions, too!
- However, do not ask or post solutions for summative (marked) assessments (4 & 5) ➡ punishable offence
- Make use of forum and the drop-in clinics; do not suffer in silence!
- I know it is tempting to use Large Language Models (like ChatGPT, Gemini, Claude etc.), but there are risks:
 - Often you require programming experience to understand LLM outputs
 - More importantly, you require experience to identify **wrong** outputs!
 - It is not allowed to use LLMs for summative (marked) assessments!

This Friday (October 3)

Pebble in the pond!



When? 11 am - 1 pm

Where?

Royal National Hotel

Next Tuesday (October 7)

Teamwork - Live Session

with Dr Fiona Truscott



When? 11 am - 1 pm

Where?

John Hanbury LT

Now / Next hour  Programming foundations!